Ex. No: 09	ACCESSING APIs THROUGH HTTP
Date of Exercise	
Git-hub Host Link	

Aim:

To access a REST API using HTTP and retrieve and display JSON data in a Flutter application.

Description:

This experiment demonstrates how Flutter communicates with a REST API using the http package. The application sends a GET request to the JSONPlaceholder API, converts the JSON response into Dart data, and displays the posts using ListView.builder. Loading and error states are also handled.

Procedure:

Step 1 : Create Flutter Project - flutter create api_app

Step 2 : Add HTTP Package - Add http under dependencies in pubspec.yaml, then run:

Step 3 : Import Packages dart:convert for JSON decoding and the http package for API requests.

Step 4 : Create API Function Create a function using http.get() to retrieve data from the JSONPlaceholder REST API.

Step 5 : Decode JSON Data check the response status and convert the JSON response into Dart data using jsonDecode().

Step 6 : Display API Data Store the retrieved posts in a list and display the title and body of each post using ListView.builder().

Step 7 : Add Loading Display a loading indicator while the API data is being fetched.

Step 8 : Display an appropriate error message when the API request is unsuccessful

Step 9 : Test the Application Run the application in Chrome and verify that the API posts are displayed correctly.

Step 10 : Deploy to GitHub Pages - Build and upload the Flutter web application to the GitHub repository exp9-api-flutter and enable GitHub Pages using GitHub Actions.

Program:

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() {
runApp(const ApiApp());
}

class ApiApp extends StatelessWidget {
const ApiApp({super.key});

@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
title: 'API App',
theme: ThemeData(
scaffoldBackgroundColor: const Color(0xFFF8F4FC),
colorScheme: ColorScheme.fromSeed(
seedColor: const Color(0xFF9C7AC7),
),
appBarTheme: const AppBarTheme(
backgroundColor: Color(0xFFBFA2DB),
foregroundColor: Colors.white,
centerTitle: true,
elevation: 2,
),
cardTheme: const CardThemeData(
color: Color(0xFFF0E6F7),
elevation: 2,
margin: EdgeInsets.symmetric(
horizontal: 12,
vertical: 7,
),
),
),
home: const ApiHomePage(),
);
}
}

class ApiHomePage extends StatefulWidget {
const ApiHomePage({super.key});

@override
```

```
State<ApiHomePage> createState() => _ApiHomePageState();
}

class _ApiHomePageState extends State<ApiHomePage> {
List<dynamic> posts = [];
bool isLoading = true;
String errorMessage = '';

@override
void initState() {
super.initState();
fetchData();
}

Future<void> fetchData() async {
setState(() {
isLoading = true;
errorMessage = '';
});

try {
final response = await http.get(
Uri.parse(
'https://jsonplaceholder.typicode.com/posts',
),
);

if (response.statusCode == 200) {
final data = jsonDecode(response.body);

setState(() {
posts = data;
isLoading = false;
});
} else {
setState(() {
errorMessage = 'Failed to load data';
isLoading = false;
});
}
} catch (e) {
setState(() {
errorMessage = 'Network error. Please try again.';
```

```
isLoading = false;
});
}
}

@override
Widget build(BuildContext context) {
return Scaffold(
appBar: AppBar(
title: const Text(
'REST API Posts',
style: TextStyle(
fontWeight: FontWeight.bold,
),
),
),

body: isLoading
? const Center(
child: CircularProgressIndicator(
color: Color(0xFF9C7AC7),
),
)
: errorMessage.isNotEmpty
? Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.center,
children: [
const Icon(
Icons.error_outline,
size: 60,
color: Color(0xFF9C7AC7),
),
const SizedBox(height: 15),

Text(
errorMessage,
style: const TextStyle(
fontSize: 18,
color: Color(0xFF5E4A6E),
),
),

const SizedBox(height: 15),
```

```
ElevatedButton(
onPressed: fetchData,
style: ElevatedButton.styleFrom(
backgroundColor:
const Color(0xFFBFA2DB),
foregroundColor: Colors.white,
),
child: const Text('Retry'),
),
],
),
)
: RefreshIndicator(
color: const Color(0xFF9C7AC7),
onRefresh: fetchData,

child: ListView.builder(
padding: const EdgeInsets.symmetric(
vertical: 10,
),
itemCount: posts.length,

itemBuilder: (context, index) {
return Card(
child: ListTile(
contentPadding:
const EdgeInsets.all(12),

leading: CircleAvatar(
backgroundColor:
const Color(0xFFD8C4E8),

child: Text(
'${posts[index]['id']}',
style: const TextStyle(
color: Color(0xFF5E4A6E),
fontWeight: FontWeight.bold,
),
),
),

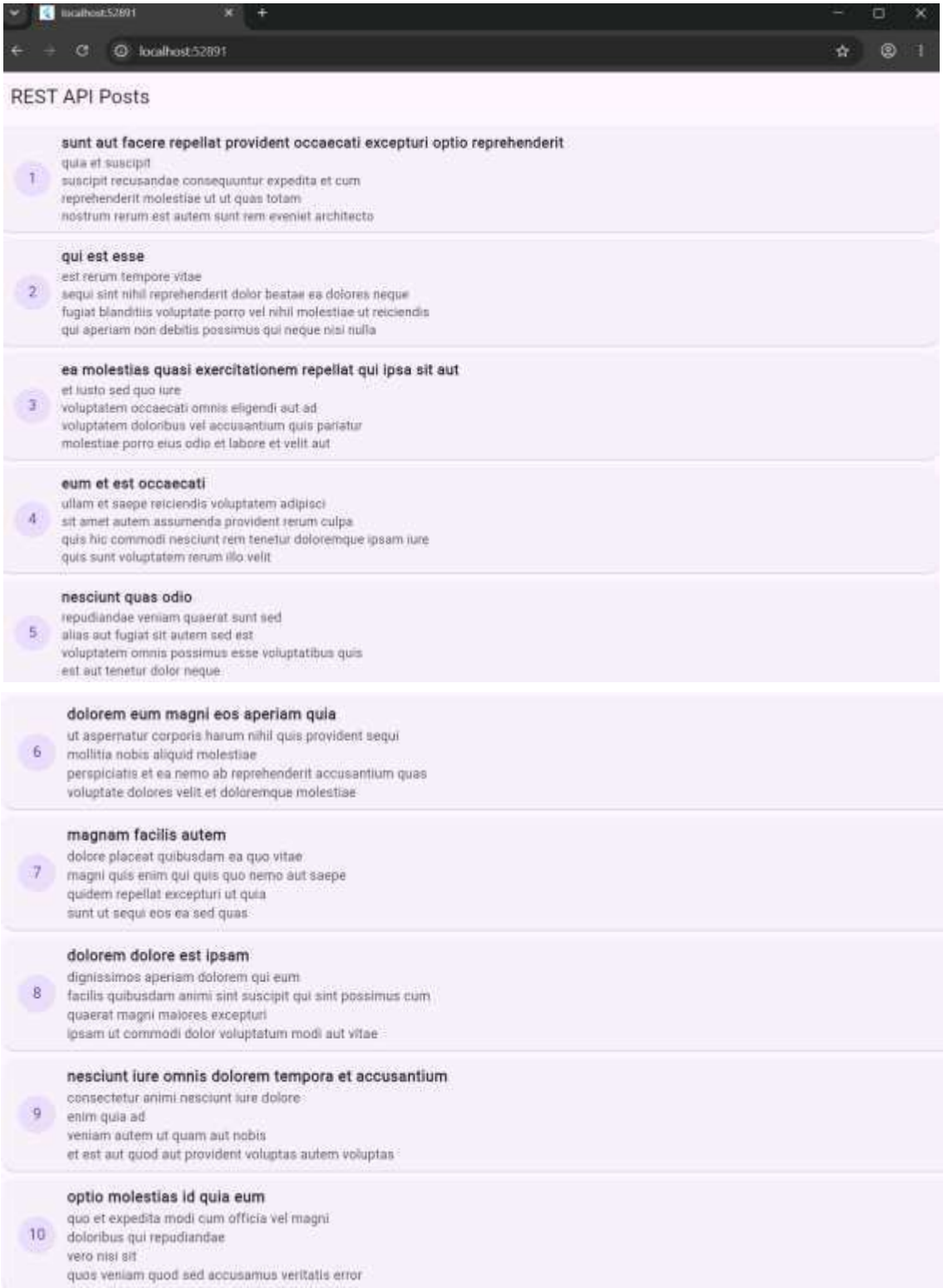
title: Text(
posts[index]['title'],
```

```
style: const TextStyle(
color: Color(0xFF4D3A5C),
fontWeight: FontWeight.bold,
fontSize: 16,
),
),

subtitle: Padding(
padding: const EdgeInsets.only(
top: 8,
),

child: Text(
posts[index]['body'],
style: const TextStyle(
color: Color(0xFF6E5A7D),
height: 1.4,
),
),
),
),
);
},
),
),
);
}
}
```

Output:


localhost:52891
localhost:52891
REST API Posts
1
sunt aut facere repellat provident occaecati excepturi optio reprehenderit
quia et suscipit
suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto
2
qui est esse
est rerum tempore vitae
sequi sint nihil reprehenderit dolor beatae ea dolores neque
fugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis
qui aperiam non debitis possimus qui neque nisi nulla
3
ea molestias quasi exercitationem repellat qui ipsa sit aut
et iusto sed quo iure
voluptatem occaecati omnis eligendi aut ad
voluptatem doloribus vel accusantium quis pariatur
molestiae porro eius odio et labore et velit aut
4
eum et est occaecati
ullam et saepe reiciendis voluptatem adipisci
sit amet autem assumenda provident rerum culpa
quis hic commodi nesciunt rem tenetur doloremque ipsam iure
quis sunt voluptatem rerum illo velit
5
nesciunt quas odio
repudiandae veniam quaerat sunt sed
alias aut fugiat sit autem sed est
voluptatem omnis possimus esse voluptatibus quis
est aut tenetur dolor neque
6
dolorem eum magni eos aperiam quia
ut aspernatur corporis harum nihil quis provident sequi
mollitia nobis aliquid molestiae
perspiciatis et ea nemo ab reprehenderit accusantium quas
voluptate dolores velit et doloremque molestiae
7
magnam facilis autem
dolore placeat quibusdam ea quo vitae
magni quis enim qui quis quo nemo aut saepe
quidem repellat excepturi ut quia
sunt ut sequi eos ea sed quas
8
dolorem dolore est ipsam
dignissimos aperiam dolorem qui eum
facilis quibusdam animi sint suscipit qui sint possimus cum
quaerat magni maiores excepturi
ipsam ut commodi dolor voluptatum modi aut vitae
9
nesciunt iure omnis dolorem tempora et accusantium
consectetur animi nesciunt iure dolore
enim quia ad
veniam autem ut quam aut nobis
et est aut quod aut provident voluptas autem voluptas
10
optio molestias id quia eum
quo et expedita modi cum officia vel magni
doloribus qui repudiandae
vero nisi sit
quos veniam quod sed accusamus veritatis error



RESULT

Thus, The Flutter application successfully accesses a REST API using the HTTP package, retrieves JSON data from JSONPlaceholder, converts the response into Dart data, and displays the posts in a scrollable UI with loading and error handling.